

14 Topology of $\mathbb{R}$ I



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 3A03 Real Analysis I

Instructor: David Earn

Lecture 14
Topology of $\mathbb{R}^1$
Monday 10 February 2025

Announcements



Topology of $\mathbb{R}$

Intervals



Open interval:

$$(a, b) = \{x : a < x < b\}$$

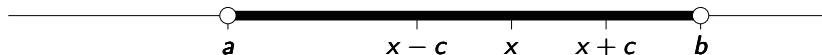
Closed interval:

$$[c, d] = \{x : c \leq x \leq d\}$$

Half-open interval:

$$(e, f] = \{x : e < x \leq f\}$$

Interior point



Definition (Interior point)

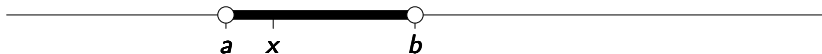
If $E \subseteq \mathbb{R}$ then x is an *interior point* of E if x lies in an open interval that is contained in E , i.e.,

$$\exists c > 0 \quad \text{such that} \quad (x - c, x + c) \subset E.$$

Interior point examples

Set E	Interior points?
$(-1, 1)$	
$[0, 1]$	
$\mathbb{N}$	
$\mathbb{R}$	
$\mathbb{Q}$	
$(-1, 1) \cup [0, 1]$	
$(-1, 1) \setminus \{\frac{1}{2}\}$	

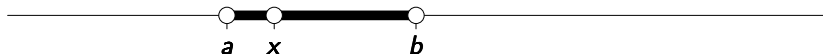
Neighbourhood



Definition (Neighbourhood)

A **neighbourhood** of a point $x \in \mathbb{R}$ is an open interval containing x .

Deleted neighbourhood

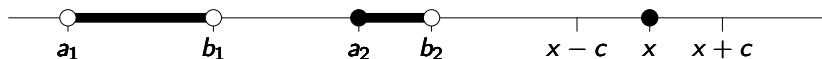


$$(a, b) \setminus \{x\}$$

Definition (Deleted neighbourhood)

A **deleted neighbourhood** of a point $x \in \mathbb{R}$ is a set formed by removing x from a neighbourhood of x .

Isolated point



$$E = (a_1, b_1) \cup [a_2, b_2] \cup \{x\}$$

Definition (Isolated point)

If $x \in E \subseteq \mathbb{R}$ then x is an **isolated point** of E if there is a neighbourhood of x for which the only point in E is x itself, i.e., $\exists c > 0$ such that $(x - c, x + c) \cap E = \{x\}$.

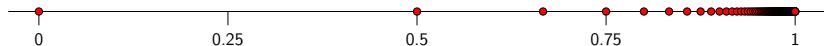
Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 3A03**
- Click on **Take Class Poll**
- Fill in poll **Topology: Isolated points**
- .

Isolated point examples

Set E	Isolated points?
$(-1, 1)$	
$[0, 1]$	
$\mathbb{N}$	
$\mathbb{R}$	
$\mathbb{Q}$	
$(-1, 1) \cup [0, 1]$	
$(-1, 1) \setminus \{\frac{1}{2}\}$	

Accumulation point



$$E = \left\{ 1 - \frac{1}{n} : n \in \mathbb{N} \right\}$$

Definition (Accumulation Point or Limit Point or Cluster Point)

If $E \subseteq \mathbb{R}$ then x is an **accumulation point** of E if every neighbourhood of x contains infinitely many points of E ,

$$\text{i.e.,} \quad \forall c > 0 \quad (x - c, x + c) \cap (E \setminus \{x\}) \neq \emptyset.$$

Note:

- It is possible but not necessary that $x \in E$.
- The shorthand condition is equivalent to saying that every deleted neighbourhood of x contains at least one point of E .

Poll

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- .

Accumulation point examples

Set E	Accumulation points?
$(-1, 1)$	
$[0, 1]$	
$\mathbb{N}$	
$\mathbb{R}$	
$\mathbb{Q}$	
$(-1, 1) \cup [0, 1]$	
$(-1, 1) \setminus \{\frac{1}{2}\}$	
$\{1 - \frac{1}{n} : n \in \mathbb{N}\}$	